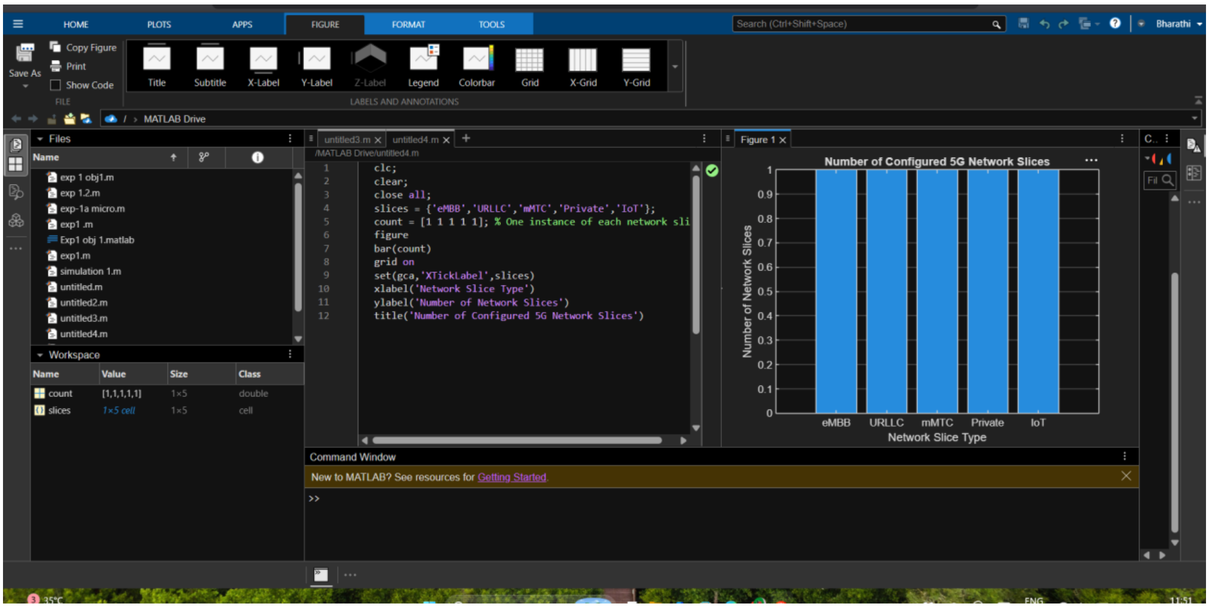
Experiment: 10: 5G Network Slicing Environment Using MATLAB

Objective – 1

```
clc;
clear;
close all;
slices = {'eMBB','URLLC','mMTC','Private','IoT'};
count = [1 1 1 1 1]; % One instance of each network slice
figure
bar(count)
grid on
set(gca,'XTickLabel',slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')
```



Objective – 2

```
clc;
clear;
```

close all;

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});

utilization = [85 70 60 75 50];

figure;

bar(slices, utilization);

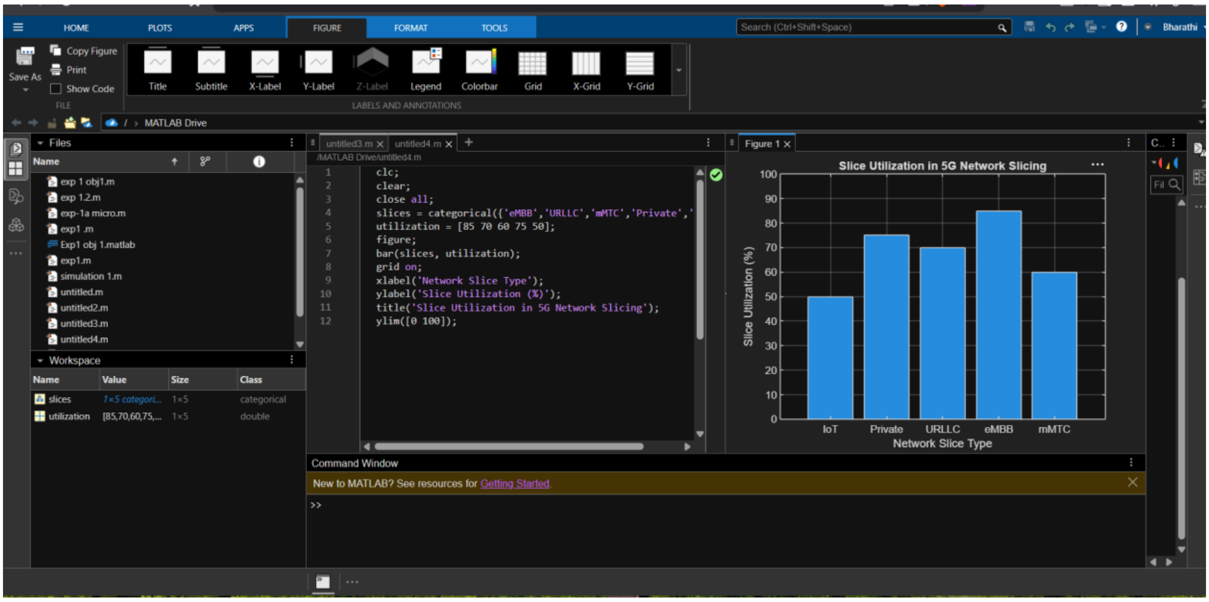
grid on;

xlabel('Network Slice Type');

ylabel('Slice Utilization (%)');

title('Slice Utilization in 5G Network Slicing');

ylim([0 100]);



Objective – 3

clc;

clear;

close all;

slices = [1 2 3 4 5];

latency = [25 5 40 15 60];

figure;

plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);

grid on;

```
xticks(slices);

xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});

xlabel('Network Slice Type');

ylabel('End-to-End Latency (ms)');

title('5G Network Slicing: End-to-End Latency');

ylim([0 70]);
```